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def ks_2d_statistic(X, Y, G):

    n = len(X)

    # observed points
    D1 = max(
        empirical_cdf(X[i], Y[i], X, Y) - G(X[i], Y[i])
        for i in range(n)
    )

    # intersection points
    D2 = max(
        empirical_cdf(X[i], Y[j], X, Y) - G(X[i], Y[j])
        for i in range(n) for j in range(n)
    )

    # lower deviation with 2/n correction
    D3 = max(
        (2.0 / n) - (
            empirical_cdf(X[i], Y[j], X, Y) - G(X[i], Y[j])
        )
        for i in range(n) for j in range(n)
        if (X[j] > X[i] and Y[j] > Y[i])
    )

    # right boundary (x = 1)
    D4 = max(
        (1/n) - (empirical_cdf(1, Y[i], X, Y) - G(1, Y[i]))
        for i in range(n)
    )

    # top boundary (y = 1)
    D5 = max(
        (1/n) - (empirical_cdf(X[i], 1, X, Y) - G(X[i], 1))
        for i in range(n)
    )

    # final statistic
    Dn = max(D1, D2, D3, D4, D5)

    return Dn

```

```
# alpha quantiles
Dn_null = []
for _ in tqdm(range(1000)):
    sample = np.random.multivariate_normal(mean0, cov, 100)
    X0, Y0 = sample[:,0], sample[:,1]
    Dn_null.append(ks_2d_statistic(X0, Y0, G_normal))
c_alpha = np.quantile(Dn_null, 0.95)
```

sample size	MC percentiles (1000 replications)	MC percentiles (paper) (10.000 replications)
15	.427	.4141
25	.33	.3254
50	.23	.2350
100	.162	.1675

```

num_sim = 500

mean0 = np.array([0,0])
cov = np.array([[1, 0.5], [0.5, 1]])

c_alpha = [0.162]
power_values = []
n = 100
epsilon_list = [0.1, 0.2, 0.4]
for epsilon in epsilon_list:
    Dn_list = []
    for k in tqdm(range(num_sim)):
        # alternative distribution
        mean1 = (3,3)
        mask = np.random.rand(n) < epsilon
        alt = np.empty((n, 2))
        alt[mask] = np.random.multivariate_normal(mean1, cov, np.sum(mask))
        alt[~mask] = np.random.multivariate_normal(mean0, cov, np.sum(~mask))
        X = alt[:,0]
        Y = alt[:,1]

        Dn_list.append(ks_2d_statistic(X, Y, G_normal))

    power = np.mean(np.array(Dn_list)>c_alpha)
    power_values.append(power)

```

empirical power of KS test (D_n) with $\alpha=0.05$: $P(D_n > c_\alpha)$

epsilon/ sample size	15	25	50	100
0.1	.096	.092	.188	.364
0.2	.236	.366	.62	.952
0.4	.758	.890	1	1

epsilon/ sample size	15	25	50	100
0.1	.13	.16	.23	.41
0.2	.27	.40	.67	.94
0.4	.73	.92	1	1